

Overview MongoDB

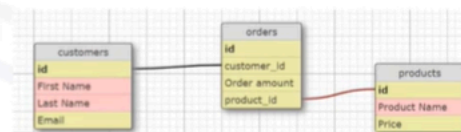
Sure! Let's dive into the concept of MongoDB, which is a type of database.

MongoDB is like a digital filing cabinet where you can store information in a flexible way. Instead of organizing data in strict rows and columns like a traditional spreadsheet, MongoDB uses documents that are similar to JSON objects or Python dictionaries. Imagine each document as a unique file that contains all the details about a specific item, like a student. For example, a student document might include their first name, last name, email, and student ID. All these documents of similar types are grouped together in collections, just like files in a folder.

This flexibility allows you to easily add or change information without needing to restructure everything, making it perfect for applications that need to handle different types of data. Plus, MongoDB can grow with your needs, whether you have a small amount of data or a massive amount. It's like having a wardrobe that can expand or shrink based on how many clothes you have!

What is MongoDB database?

- A document and a NoSQL database



- Where data is structured in non-relational way



Order document

- MongoDB is a document and a NoSQL database. Instead of storing data in tables of rows or columns like SQL databases, each record in a MongoDB database is a document, and you store the data in a non-relational way.

What are documents?

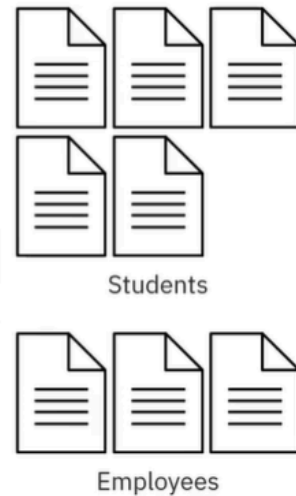
- Associative arrays like JSON objects or Python dictionaries
- For example: A student document

```
{  
  "firstName": "John",  
  "lastName": "Doe",  
  "email": "john.doe@email.com",  
  "studentId": 20217484  
}
```

- Documents are associative arrays like JSON objects or Python dictionaries. For example, a student document representing a student's first name, last name, email address, and ID.

What is a collection?

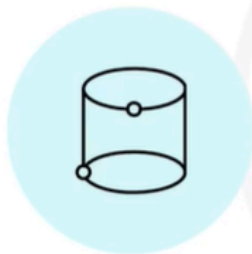
- Is a group of stored documents
- For example, all student records in Students section (**collection**) and
- Staff records in Employees section (**collection**)



- MongoDB documents of similar type are grouped into a collection. Our Campus Management system stores all student records or documents in the student's collection, and similarly all staff documents are stored in the employees collection.

What is a database?

- A database stores collections



Students and Employees collections
stored in a database called
CampusManagementDB

- So, what is a database? Following our previous example, the Campus Management system stores all different types of data related to it in a MongoDB database called CampusManagementDB. Let's dissect our document.

Documents in detail - 1/2

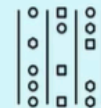
See the following fields in the document:
firstName, lastName, email and studentId

```
{
  "firstName": "John",
  "lastName": "Doe",
  "email": "john.doe@email.com",
  "studentId": 20217484
}
```

Documents in detail - 2/2

MongoDB supports various data types:

```
{
  "name": "John",
  "dateOfBirth": ISODate("2000-01-01T14:45:00.000Z"),
  "studentId": 20217484,
  "enrolled": true,
  "balance": 20.01,
  "address": {
    "city": "Stonefield",
    "country": "UK"
  },
  "interests": ["football", "skiing", "travelling"]
}
```



- In the document below, first name, last name, email, and student ID are fields or properties with values representing a student named John Doe, and each field name is unique within that document. MongoDB supports a variety of data types, so you have the full capability to use the correct data type to store your information in.
- For example, dates should be stored as ISODate or Unix-style epoch dates, which will help you with queries like, give me all students born between 15th January to 15th February. Numbers can be stored either as whole numbers or decimals. Being a document database, MongoDB also allows you to store subdocuments to group secondary information together. It also supports lists of values, not just text, but even different kinds mixed together.

Why use MongoDB?

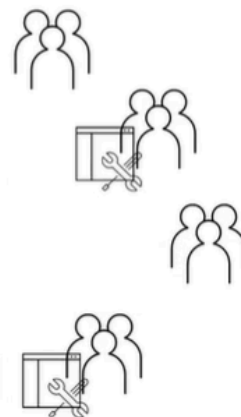
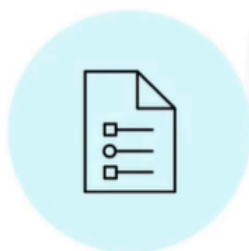
- Model data as you read/write, not the other way
 - Traditional relational databases: Create the schema first, then create the tables
 - To store another field, you have to alter tables

```
SELECT * FROM Orders INNER JOIN Customers...  
ALTER TABLE Customers...
```



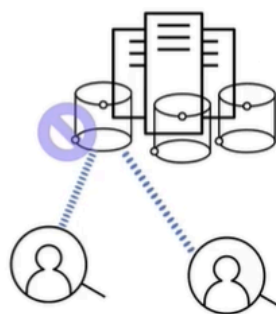
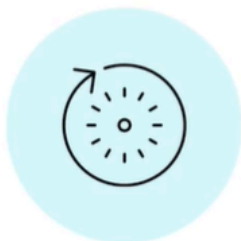
Why use MongoDB?

- Model data as you read/write, not the other way
- Bring structured/unstructured data



Why use MongoDB?

- Model data as you read/write, not the other way
- Bring structured/unstructured data
- High availability



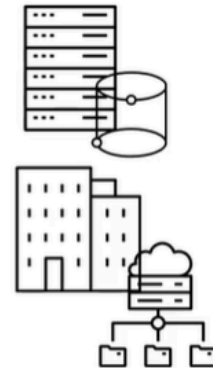
- Working with MongoDB is easy because you can focus on the data you are writing and how you are going to read it. Traditional relational databases required you to create the schema first, then create table structures that will hold your data. So, if you decide to store an additional field, you will then have to alter your tables. In MongoDB, you change as you go along. MongoDB also

gives you the power to bring in any structured or unstructured data. It also provides high availability by keeping multiple copies of your data.

Where to use MongoDB

MongoDB is a popular choice of database for

- Large and unstructured
- Complex
- Flexible
- Highly scalable applications
- Self-managed, hybrid, or cloud hosted



<https://www.ibm.com/cloud/databases-for-mongodb>

- You can design complex data structures easily in MongoDB without worrying about the complexity of how it is stored and how it should be linked. For example, your Campus Management application is also launched in the USA, where they do not store postcodes but have zip codes.
- Scalability provided by MongoDB means as your data needs grow, you can easily scale vertically by introducing bigger, faster, better hardware, or horizontally by partitioning your data. All of that can be done whether you are running a self-managed on-premises MongoDB or the hybrid or cloud-hosted and fully managed services such as IBM Cloud databases for MongoDB or MongoDB Atlas on AWS, Azure, and Google Cloud.

Recap

In this video, you learned that:

- MongoDB is a document and a NoSQL database
- MongoDB supports various data types
- Documents provide a flexible way to store data
- MongoDB documents of similar type are grouped into collections
- MongoDB models data as you read/write, brings structured or unstructured data, and provides high availability
- MongoDB can be used for a variety of purposes